

Foundations of CLIP neural network

[CLIP neural network]

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

1.0 Content Block

resorting to entirely biologically inspired local learning rules for the hidden units translating conventionally trained "rate-based" NNs to SNNs smoothing the network model to be continuously differentiable defining an SG (Surrogate Gradient) as a continuous relaxation of the real gradients The second concerns the optimization algorithm. Standard BP can be expensive in terms of computation, memory, and communication and may be poorly suited to the hardware that implements it (e.g., a computer, brain, or neuromorphic device). Incorporating additional neuron dynamics such as Spike Frequency Adaptation (SFA) is a notable advance, enhancing efficiency and computational power. These neurons sit between biological complexity and computational complexity. Originating from biological insights, SFA offers significant computational benefits by reducing power usage, especially in cases of repetitive or intense stimuli. This adaptation improves signal/noise clarity and introduces an elementary short-term memory at the neuron level, which in turn, improves accuracy and efficiency. This was mostly achieved using compartmental neuron models. The simpler versions are of neuron models with adaptive thresholds, are an indirect way of achieving SFA. It equips SNNs with improved learning capabilities, even with constrained synaptic plasticity, and elevates computational efficiency. This feature lessens the demand on network layers by decreasing the need for spike processing, thus lowering computational load and memory access time—essential aspects of neural computation. Moreover, SNNs utilizing neurons capable of SFA achieve levels of accuracy that rival those of conventional ANNs, while also requiring fewer neurons for comparable tasks. This efficiency streamlines the computational workflow and conserves space and energy, while maintaining technical integrity. High-performance deep spiking neural networks can operate with 0.3 spikes per neuron.

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Information in the brain is represented as action potentials (neuron spikes), which may group into spike trains or coordinated waves. A fundamental question of neuroscience is to determine whether neurons communicate by a rate or temporal code. Temporal coding implies that a single spiking neuron can replace hundreds of hidden units on a conventional neural net. SNNs define a neuron's current state as its potential (possibly modeled as a differential equation). An input pulse causes the potential to rise and then gradually decline. Encoding schemes can interpret these pulse sequences as a number, considering pulse frequency and pulse interval. Using the precise time of pulse occurrence, a neural network can consider more information and offer better computing properties. SNNs compute in the continuous domain. Such neurons test for activation only when their potentials reach a certain value. When a neuron is activated, it produces a signal that is passed to connected neurons, accordingly raising or lowering their potentials. The SNN approach produces a continuous output instead of the binary output of traditional ANNs. Pulse trains are not easily interpretable, hence the need for encoding schemes. However, a pulse train representation may be more suited for processing spatiotemporal data (or real-world sensory data classification). SNNs connect neurons only to nearby neurons so that they process input blocks separately (similar to CNN using filters). They consider time by encoding information as pulse trains so as not to lose information. This avoids the complexity of a recurrent neural network (RNN). Impulse neurons are more powerful computational units than traditional artificial neurons. SNNs are theoretically more powerful than

so called "second-generation networks" defined as ANNs "based on computational units that apply activation function with a continuous set of possible output values to a weighted sum (or polynomial) of the inputs"; however, SNN training issues and hardware requirements limit their use. Although unsupervised biologically inspired learning methods are available such as Hebbian learning and STDP, no effective supervised training method is suitable for SNNs that can provide better performance than second-generation networks. Spike-based activation of SNNs is not differentiable, thus gradient descent-based backpropagation (BP) is not available. SNNs have much larger computational costs for simulating realistic neural models than traditional ANNs. Pulse-coupled neural networks (PCNN) are often confused with SNNs. A PCNN can be seen as a kind of SNN. Researchers are actively working on various topics. The first concerns differentiability. The expressions for both the forward- and backward-learning methods contain the derivative of the neural activation function which is not differentiable because a neuron's output is either 1 when it spikes, and 0 otherwise. This all-or-nothing behavior disrupts gradients and makes these neurons unsuitable for gradient-based optimization. Approaches to resolving it include:

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Spiking neural networks (SNNs) are artificial neural networks (ANN) that mimic natural neural networks. These models leverage timing of discrete spikes as the main information carrier. In addition to neuronal and synaptic state, SNNs incorporate the concept of time into their operating model. The idea is that neurons in the SNN do not transmit information at each propagation cycle (as it happens with typical multi-layer perceptron networks), but rather transmit information only when a membrane potential—an intrinsic quality of the neuron related to its membrane electrical charge—reaches a specific value, called the threshold. When the membrane potential reaches the threshold, the neuron fires, and generates a signal that travels to other neurons which, in turn, increase or decrease their potentials in response to this signal. A neuron model that fires at the moment of threshold crossing is also called a spiking neuron model. While spike rates can be considered the analogue of the variable output of a traditional ANN, neurobiology research indicated that high speed processing cannot be performed solely through a rate-based scheme. For example humans can perform an image recognition task requiring no more than 10ms of processing time per neuron through the successive layers (going from the retina to the temporal lobe). This time window is too short for rate-based encoding. The precise spike timings in a small set of spiking neurons also has a higher information coding capacity compared with a rate-based approach. The most prominent spiking neuron model is the leaky integrate-and-fire model. In that model, the momentary activation level (modeled as a differential equation) is normally considered to be the neuron's state, with incoming spikes pushing this value higher or lower, until the state eventually either decays or—if the firing threshold is reached—the neuron fires. After firing, the state variable is reset to a lower value. Various decoding methods exist for interpreting the outgoing spike train as a real-value number, relying on either the frequency of spikes (rate-code), the time-to-first-spike after stimulation, or the interval between spikes.